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ITSM Project

IT Service Management

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Academic Year 2025-2026

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Abstract

The recent international expansion of the Bank through the acquisition of other financial entities represents an important opportunity to increase the value of the services provided and achieve several business objectives. However, this expansion also introduces new challenges related to service integration, operational efficiency, regulatory compliance, and the consistent delivery of high-quality IT services across different countries and business units.

By analyzing the IT Service Management (ITSM) environment of the Bank presented in the case study, this project supports the strategic transformation of IT Department into a "Shared Service Center" thus to operate uniformly, while providing high-quality services across different countries and business units.

Moreover, to achieve business objectives, this document analyzes the individuated Critical Success Factors (CSFs): Customer Partnership, Process Factory, Continuous Improvement. Moreover through this document defines different Key Performance Indicators (KPIs), providing a framework template with their respective metric and process involved for the areas identified.

Chapter 1

Organisational Mission and Goals

1.1 Mission

The Bank's cuyrrent state is composed by various levels divided into divisions, departments and operational units, as well represented into the organization organigram in Figure 1.1. The recent acquisition of overseas banks has positioned the Bank at an international level, changing its requirements and creating new opportunities to expand and enhance the services provided. To ensure the realization of the vision, Information Technology (IT) Department cover a strategic role, by adopting the structure of a **"Shared Service Center"** thus to ensure that business services remains reliable, secure and efficient; while providing high-performance, high-quality, and cost-effective services both internally than externally to the organization.

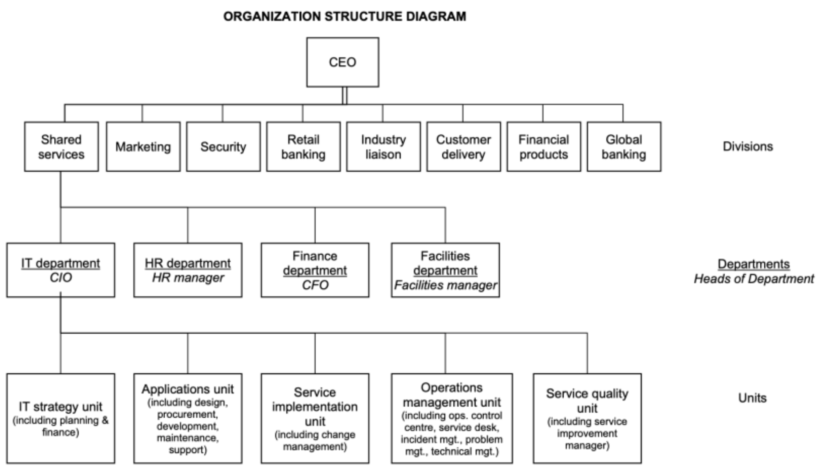

Figure 1.1: Organization Organigram

1.2 Goals

To support the Mission of the company, the primary goal of the IT Department is to implement a "Performance-Based Management" approach. Through this model, organizational performance is continuously monitored via measurable Key Performance Indicators (KPIs), which provide objective information for decision-making *(progress iteratively with feedback)*. Decisions based on measurable data, rather than subjective opinions enable management to identify weaknesses and monitor progress toward strategic objectives; thus to evaluate the effectiveness of implemented improvements and continuously enhance service quality.

To achieve these strategic objectives, the Bank has identified three Critical Success Factors (CSFs), representing the key areas that require enforcement and attention.

- **Customer Partnership** represnets the first Critical Success Factor. As the Bank expands its international presence and strengthens collaborations with external financial institutions and independent financial advisors, establishing strong relationships with customers and business partners becomes essential *(collaborate and promote visibility)*. The IT Department therefore works closely with business units to actively involve stakeholders throughout the service lifecycle. Through continuous communication, customer feedback and satisfaction surveys, IT can better understand business needs and deliver services that enrich value for customers while supporting the Bank's strategic business. As indicated by the corporation and business strategy by the bank:
  - In the bank's HQ country, to have as a customer base 10% of the retail banking market (currently 7 %)
  - To increase income from internet banking by 70
  - To increase the income from customers living outside the HQ country by 50%
  - To achieve a 5% increase in corporate profit.
- The second Critical Success Factor is **Process Factory**. The Bank aims to standardize current IT processes, technologies, and operational procedures across all acquired organizations and various departments *(start where you are)*. Standardization reduces complexity, minimizes waste, simplifies maintenance activities and improves service consistency.
  Furthermore, standardized processes facilitate the integration of different IT environments and create a clearer workflow of the process, enabiling a more efficeint management of services on an international scale *(keep it simple and practical)*. This objective is particularly important and fragile, since the Bank while maintaining high levels of service quality in multiple countires must ensure different regulations compliance of various nature: Political, Economical, Social, Technological, Environmental and Legal (PESTLE).
- The third Critical Success Factor is **Continuous Improvement**. The Bank recognizes that delivering high-quality IT services requires continuous measurement, evaluation and optimization toward to improve the value provided *(focus on value)*.
  This initiative must be done by collecting and analyzing performance data generated through KPIs; then the corporation can identify opportunities for improvement and monitor potential risks. Overral through incremental improvements, regular feedback, and continuous evaluation of results, the organization can steadily increase the value delivered to customers while optimizing both services and internal resources *(think and work holistically)*.

To verify whether these Critical Success Factors are being achieved, the Bank has defined a KPI tree composed of six Key Performance Indicators.

Together, these KPIs provide a balanced overview of service quality, operational efficiency, customer satisfaction and financial performance. By continuously monitoring these indicators, and where possible, automating the collection and analysis of performances *(optimize and automate)*, the IT Department can ensure that its activities be aligned with overall business strategy. Furthermore, KPI monitoring supports Service Level Management, enabling the organization to identify areas that require corrective actions and ensure that the service provided accomplish the customer expectations. Moreover, all these procedures ensure the continuous progress toward its strategic objectives *(keep momentum going)*.

Figure 1.2: what to do

Chapter 2

Critical Success Factors

2.1 Customer Partnership

Customer Partnership represents an important and strategic pillar to accomplish the Bank's goals. The IT Department must establish measurable relationships between external customers and other stakeholders, assuming a role that actively contributes to the improvement of the services provided and, in specific, monitors through KPIs the value perceived by customers. The IT has to collaborate also internally within the organization in order to engage the appropriate department whenever it is needed.

This topic is fundamental for the success of the Bank's vision, because the international expansion through acquisitions and other partnerships introduces new business requirements due to different customer expectations, cultural diversity, and increased service complexity. These challenges must be supported in a proactive way, by ensuring that IT services remain aligned with business objectives, deliver consistent value across different countries, and contribute to the Bank's growth, while complying with agreed Service Level Agreements between customers, stakeholders and user expectations.

These requisites, makes performance measurement crucial. By continuously monitoring and collecting customer-oriented metrics and analyzing services scores, IT Department is able to implement a continual service improvements that can strengthen customer trust and enrich the service quality. Gaining long-term partneship. This customer-centric approach supports organization's objectives of delivering high-performance, high-quality and cost-effective IT services. Aimed toward to accomplish the predefined goals **??**

2.1.1 Time To Market

Time to market, also known as Speed to Market is made operational through **Average Lead Time for Change to Production** KPI metric, defines as the elapsed time between formal approval of a business requirement and its deployment into the production environment. This metric is selected because it directly reflects the organization's ability to translate customer and business demands into operational banking capabilities which is essential for achieving the stated rapid digital service evolution and revenue growth objectives of the organisation. In the context of the strategic goal of the organisation is to increase the income from internet banking, reduced Time to Market provides higher organizational agility and faster delivery of customer targeting features, thus improving market responsiveness and competitive positioning.

There are several ITIL aligned ITSM processes that primarily effects Time to Market. **Change enablement** has a direct effect through approval workflows and Risk Management procedures, were excessive governance overhead can result in significant delays.
Release and Deployment Management also effects the indicator as deployment schedul-

ing , release batching and automation maturity determine how quickly the validated changes end up deployed in the production.
Service Validation and Testing influences Lead Time through test cycle duration and error handling loops, which can create bottlenecks if automation coverage is insufficient.
Service Design and Transition activities also effect indirectly throught the time required for requirement analysis, architectural validation, and coordination of technical dependencies. Overall, inefficiencies or lack of cooperation across these processes will result in the increase of Lead Time, where higher process integration, standardization and automation would reduce it, resulting with the improvement of Time To Market KPI.

2.1.2 Performance To Contract

Performance to Contract focuses on the IT Department‘s ability to respect contractual obligations and consistently deliver services according to the agreements made with customers and explained in natural language in the SLAs.

This subject requires IT services to meet the agreed levels of different measurable values, such as availability, reliability, responsiveness, and quality, while ensuring and supporting the Bank's operational and predefined business objectives.

Well-defined Key Performance Indicators ensure evidence of whether service targets are being achieved and enable the IT department to identify eventual deviations or deficiencies. This gives a clear input to potential corrective actions or improvements to service performance over time. Moreover, the monitoring of performance against contractual commitments also increases transparency and accountability between IT and different business units or external third-party service providers.

This process is fundamental and very important for the Bank, because many banking services, such as Online Banking, ATMs, global payments, or other third-party services, depend strictly on service levels and regulatory compliance. Also, through a cybersecurity point of view, the respect of performance is fundamental to prevent various cyber-attacks and potential disservices.
In addition, being a global Bank, the services must be ensured not only during the local time of the main Headquarters, but the support must be available respecting different cultural scenarios, thus supporting suppliers or international partners.

As a result of continuously measuring contractual performance and acting on the collected data, IT Department can improve service reliability, reduce the risk of SLA breaches and improve the overall customer relationships and partnerships with various stakeholders.

2.2 Process Factory

The Process Factory CSF approach is required in order to transform the IT department into a Shared Service Center, where IT Services are delivered through standardized, repeatable, and performance-oriented processes. This CSF is important considering the bank's international expansion strategy, the integration of acquired oversees banks and the consolidation of support functions within the Shared Services Division.

The Process Factory approach aims to standardize IT Service Management (ITSM) processes in order to deliver services with greater efficiency, higher quality, and lower operational costs. Within this approach, one of the primary objectives is the systematic identification and elimination of waste, defined as any activity that consumes resources without creating value for customers or supporting business objectives. Moreover the standardization of process helps the creation of waste and redundant steps or activities....to be continued/reviewed

2.2.1 Standardization

2.2.2 Waste

With the term **waste** IT Service Management intend every activioty that consumes time, effort or financial resources without increasing the value delivered to the customer. From Lean methodology and ITIL perspective, customers only perceive value when an IT service is available, reliable, secure and supports the desired utility functions. Any process that do not contribute to outcomes is considered waste and should be minimized or eliminated (*as the guiding principles suggests: "Keep it simple and practical"*).

Waste includes more than obvious failures, such as services outages but also hidden inefficiencies embedded in recurrent operations. These may include redudant processes steps, unnecessary approvals, excessive manual work, repeated incident resolution, but even the collections of data taht is never used to improve services.
Such activities increase opreational costs, reducing productivity and delaying service delivery, impact negatively ithe customer satisfaction.

Lean thinking identifies different categories of waste that can be directly applied to IT Service Management, such as:

- **Waiting**: delays caused by approvals, unavailable personnel or dependencies between teams. *Waiting increases lead time without adding value.*
- **Defects**: errors that require correction, like failed software releases, configuration mistakes, security vulnerabilities, or inaccurate service records.
- **Overprocessing**: performing more work than necessary, including excessive documentation not required or needed by stakeholders. Even unnecessary or redundant approvals, or too many resources allocated to simple activities (often wrongly added near to a delivery deadline).
- **Unused talents**: failing to involve employees with valuable expertise in process improvement, service design or decision-making. *Opposing the "Collaborate and promote visibility" guiding principle.*
- **Inventory**: backlogs of unresolved incidents, accumulated change requests, unused hardware, and outdated or unused retreived information.

These concepts can be applied through concrete examples to the Bank's use case scenario. For instance, without a structured **knowledge management** that holds documentation and reusable knowledge, service desk analysts and technical staff repeatedly solve the same problems independently. This duplication of effort represents both overprocessing and inventory waste, increasing incident resolution times and reducing overall efficiency.
Another source of waste comes from poor involvement of internal customers during the Service Transition and Service Design, as a result a lot of misunderstandings and usability issues could emerge that converge into rework, additional testing, and corrective changes. This represents both waiting and defects of a service, since problems are identified later in the service lifecycle or already in a production environment

An important aspect that the Bank needs to handle, thus avoiding waste, is due to the expansion into different countries. Social aspects such as language, culture, time zones, and organizational practices could end up creating a bottleneck for the communication system, delaying the throughput of information and slowing incident resolution, change approvals, or decision-making.

Even the differences in operational practices, in the absence of standardized processes 2.2.1, could represent a duplication of work and the creation of organizational silos, which can slow down decisions taken toward goal realization, representing the opposite of one of *ITIL core guiding principle "Think and work holistically"*. For the Bank, reducing waste is a necessity that allows IT resources to focus on innovation, improvements in service quality and lower operating costs. In order to individuate waste, the organization must ensure standardization of processes, automate repetitive activities, define responsibilities and continuously monitor performance through KPIs (analyzed in the following chapter ??). This transformation of strategy makes the IT department operate as a "Shared Service Center" that provides high performance, high quality and cost-effectiveness, aligned with the organization's goals.

2.3 Continuous Improvement

2.3.1 Service Costs

2.3.2 Backlog

Backlog represnets a structured and prioritized repository of improvements initiatives taken or identified through service performance measurement methods like KPI monitoring, incidents, problem analysis or customer feedback and reviews. Provide a governance mechanism that enables the organization to systematically collect, evaluate, pripritize and monitor the improvement process and opportunities in a shared system rather than operate as an organizational silos. In fact, backlog represent a key tool for the transformation of the IT Department into a "Shared Service Center", because improvement initiatives are continuously generated from the monitoring of KPI across various processes. The adoption of a welformed backlog gives the IT the possibility to provide and suggest different initiatives thus to ensure the investments of resource and operation that contribute directly to the organization's goals and support informed decision-making.

Initiatives are prioritized by different factors, such as:

- **Business value** that provide to the customers
- **Strategic alignment** toward organisation's goals
- **Implementation effort**
- **Operational risk**
- **Expected benefits**
- **Resouce ability**

The backlog must be continuously reviewed as part of the *Continual Service Improvement* process, allowing new opportunities to be integrated while completed activities can be evaluated against predefined performance indicators. The outcome of each implemented improvement is measured through opportune KPIs to verify and then support the validation of the achievement of the expected benefits. Moreover, can serve to identify additional optimization and creating a feedback loop that promotes organizational learning and maturity.

Applied to the Bank current ITSM environment, the adoption of an improvement backlog brings several benefits to the services provided, such as:

- Enrich and consolidate a Knowledge Management
- Increase customer involvement during Service Transition
- Introduce formal change evaluation process
- Promote proactive Problem Management
- Standardize operational process across international branches and improve collaboration between departments

Together, these initiatives contribute the reducing of operational waste while improving service quality, increasing process efficiency and supporting Banks's international growth strategy through a culture of continuous improvement.

Chapter 3

Measurement Procedures

3.1 Time to Market

3.2 Performance to Contract

Explained in subsection 2.1.2.

3.2.1 SLA Compliance Rate

SLA Compliance Rate can provide a measure whether contractual service levels are respected, thus to accomplish the agreements made with customers or stakeholders and therefore maintaint the quality of the service provided.

For this KPI different process are affected, the first is the "Service Level Management", "Incident Management", Request

For this KPI the process affected are:

- **Service Level Management**: to ensure an opportune level of service provided that should ensure what agreed in SLA
- **Incident Management**: to ensure that eventual incident is correctly resolved within time and quality specified.
- **Request Fulfillment**: to fulfill customer and user requests according the agreed specifications
- **Availabiliy Management**: to ensure the service is available for the user and the downtime dued to incidents or maintenance do not take more than necessary resources (time and costs)
- **Capacity Management**: to monitor that the service fulfill the capacity and performance designed.
- **Supplier Management**: whenever a process involves suppliers and third part activities not strictly dependent by the organization.

SLA Compliance Rate	
Description	**Unit**
Percentage of request solved respecting SLAs specification, where is desired a high percentage.	Percentage.
Formula	**Frequency**
$$\frac{\text{Requests solved within SLA}}{\text{Total requests}} \times 100$$	Monthly.

Table 3.1: Measurement - SLA Compliance Rate

3.2.2 Mean restoring service time

3.3 Standardization

3.4 Waste

3.4.1 Repeated Incident rate

To measure how many incident occur dued to problems that are not already been resolved. ITSM process involved are

- **Incident Management** which is accountable to register the incident and potential workaround used in restoring the service
- **Problem Management** which is responsable to resolve the problem that cause the incident or the defect of the service
- **Knowledge Management** thus to inform the organization of known errors
- **Continual Improvement** in order to increase the value overral the performance and quality of the service

Repeated Incident Rate	
Description	**Unit**
A high value indicate a wasted effort since the cause of the incident has not been eliminated.	Percentage.
Formula	**Frequency**
$$\frac{\text{Recurring incidents}}{\text{Total incidents}} \times 100$$	Weekly/Monthly.

Table 3.2: Waste - Repeated Incident Rate

3.4.2 Rework rate

For instance, since the Bank aims to increase revenue from its Internet Banking services, poor user experience caused by application malfunctions or insufficient service quality may reduce customer engagement. At the same time, these defects can represent significant rework for both external suppliers and the internal design and development teams; increasing costs and delaying service improvements, reducing the overall efficiency of the Service Value Chain.

The rate of the rework activities serve a strong KPI for user engagement and involves many process of IT Service Management, main of them are:

- **Change Enablement**: since every change request must be valuated and analyzed.
- **Release Management**: a rework require a further deployment of the new version of the service.
- **Service Validation and Testing**: in order to check that the service meets the desired customer requests.

Rework Rate	
Description	**Unit**
indicate how many activities and work item needs a rework. Higher value indicated inefficiency and waste.	Percentage.
Formula	**Frequency**
$$\frac{\text{Work task reopened}}{\text{Total completed work task}} \times 100$$	Montly.

Table 3.3: Waste - Rework Rate

3.5 Service Costs

3.6 Backlog

Number of open incidents, problems, service requests and change requests awaiting completion | decrease

Exists several backlog solutions to implement thus to resolve problems like the number of open incidents, problems, service requests and change requests awaiting approval., thus to resolve problems explained in the respective section 2.3.2

Number of open incident	
Description	**Unit**
Number of unresolved incidents older than SLA.	Number of tickets.
Formula	**Frequency**
$$\sum_{\substack{i=0 \\ \text{ticket}_i=\text{open} \\ \text{time}_i>\text{SLA}}}^{n} x_i$$ With "n" as total number of ticket within the frequency preferred.	Daily/Weekly/Montly.

Table 3.4: Measurement - SLA Compliance Rate

Chapter 4

Conclusions

whats the impact of our work for the company

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